

Carnet: 00225424

A: 00

B: 24

Red base asignada:

Ejercicio 1: VLSM y enrutamiento estático en GNS3 (2 puntos) Utilizando la red base personalizada 10.B.A.0/24, diseñe e implemente el direccionamiento IP de la topología indicada por el docente. Posteriormente, configure rutas estáticas para permitir comunicación entre todas las redes 10.24.0.

- Pruebas de ping exitosas entre PC1, PC2, PC3 y PC4.

```
PC1> ip 10.24.0.2 255.255.255.192 10.24.0.1
Checking for duplicate address...
savePC1 : 10.24.0.2 255.255.255.192 gateway 10.24.0.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> ping 10.24.0.98
10.24.0.98 icmp_seq=1 timeout
34 bytes from 10.24.0.98 icmp_seq=2 ttl=62 time=60.792 ms
34 bytes from 10.24.0.98 icmp_seq=3 ttl=62 time=61.359 ms
34 bytes from 10.24.0.98 icmp_seq=4 ttl=62 time=62.728 ms
34 bytes from 10.24.0.98 icmp_seq=5 ttl=62 time=60.563 ms

PC1> ping 10.24.0.114
10.24.0.114 icmp_seq=1 timeout
34 bytes from 10.24.0.114 icmp_seq=2 ttl=61 time=91.430 ms
34 bytes from 10.24.0.114 icmp_seq=3 ttl=61 time=92.667 ms
34 bytes from 10.24.0.114 icmp_seq=4 ttl=61 time=91.459 ms
34 bytes from 10.24.0.114 icmp_seq=5 ttl=61 time=92.462 ms

PC1> ping 10.24.0.66
10.24.0.66 icmp_seq=1 timeout
34 bytes from 10.24.0.66 icmp_seq=2 ttl=63 time=31.098 ms
34 bytes from 10.24.0.66 icmp_seq=3 ttl=63 time=30.742 ms
34 bytes from 10.24.0.66 icmp_seq=4 ttl=63 time=30.450 ms
34 bytes from 10.24.0.66 icmp_seq=5 ttl=63 time=30.316 ms

PC1> trace 10.24.0.114
Trace to 10.24.0.114, 8 hops max, press Ctrl+C to stop
 1  10.24.0.1  15.791 ms  15.030 ms  15.572 ms
 2  10.24.0.129 45.477 ms 46.270 ms 46.817 ms
 3  10.24.0.133 77.760 ms 75.838 ms 75.785 ms
 4  * * *
 5  *10.24.0.114 91.813 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>
```

- Prueba de conexiones entre routers

```

PC1 PC2 PC3 PC4 R1 R2 R3
*Mar 1 00:03:24.051: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:25.051: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config)#interface serial2/0
R1(config-if)#ip address 10.24.0.130 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:03:59.967: %LINK-3-UPDOWN: Interface Serial2/0, changed state to up
R1(config)#
*Mar 1 00:04:00.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R1(config)#
*Mar 1 00:04:22.291: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down
R1(config)#
*Mar 1 00:05:52.303: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R1(config)#ip route 10.24.0.96 255.255.255.240 10.24.0.129
R1(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.129
R1(config)#ip route 10.24.0.132 255.255.255.252 10.24.0.129
R1(config)#exit
R1#
*Mar 1 00:22:32.975: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C       10.24.0.0/26 is directly connected, FastEthernet0/0
C       10.24.0.64/27 is directly connected, FastEthernet0/1
S       10.24.0.112/28 [1/0] via 10.24.0.129
S       10.24.0.96/28 [1/0] via 10.24.0.129
C       10.24.0.128/30 is directly connected, Serial2/0
S       10.24.0.132/30 [1/0] via 10.24.0.129
R1#
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```

```

PC1 PC2 PC3 PC4 R1 R2 R3
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface Serial2/1
R2(config-if)#ip address 10.24.0.134 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:05:52.287: %LINK-3-UPDOWN: Interface Serial2/1, changed state to up
R2(config)#
*Mar 1 00:05:53.291: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/1, changed state to up
R2(config)#
*Mar 1 00:06:22.499: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/1, changed state to down
R2(config)#
*Mar 1 00:07:32.499: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/1, changed state to up
R2(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.130
R2(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.130
R2(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.133
R2(config)#exit
R2#
*Mar 1 00:21:53.763: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.130
S       10.24.0.64/27 [1/0] via 10.24.0.130
S       10.24.0.112/28 [1/0] via 10.24.0.133
C       10.24.0.96/28 is directly connected, FastEthernet0/0
C       10.24.0.128/30 is directly connected, Serial2/0
C       10.24.0.132/30 is directly connected, Serial2/1
R2#
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```

```
PC1 PC2 PC3 PC4 R1 R2 R3
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#ip address 10.24.0.113 255.255.255.240
Mar 1 00:05:19.235: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
Mar 1 00:05:20.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#interface serial12/0
R3(config-if)#ip address 10.24.0.133 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
Mar 1 00:05:46.347: %LINK-3-UPDOWN: Interface Serial12/0, changed state to up
R3(config)#
Mar 1 00:05:47.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial12/0, changed state to up
R3(config)#ip route 10.24.0.2 255.255.255.192 10.24.0.134
Inconsistent address and mask
R3(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.134
R3(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.134
R3(config)#ip route 10.24.0.128 255.255.255.252 10.24.0.134
R3(config)#exit
R3#show
Mar 1 00:20:34.731: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
    10.24.0.0/26 [1/0] via 10.24.0.134
    10.24.0.64/27 [1/0] via 10.24.0.134
    10.24.0.112/28 is directly connected, FastEthernet0/0
    10.24.0.128/30 [1/0] via 10.24.0.134
    10.24.0.132/30 is directly connected, Serial12/0
R3#
```

Tabla de subneteo VLSM completa. Dirección de red, máscara, primer host, último host y broadcast.

00.22.54.24

Ad = 24 = 64 = 256
 V = 23 = 32 = 271
 S = 24 = 16 = 281
 G = 24 = 16 = 281
 R = 22 = 4 = 301
 F = 24 = 4 = 301

A = 00 10.00.24.0/24
 B = 24

Area	Host	IP	1P Util	ultima IP	Mascara	Broadcast
Admin	45	10.24.0.0	10.24.0.1	10.24.0.62	26/	10.24.0.63
Ventas	30	10.24.0.64	.65	.94	27/	.95
Soporte	14	10.24.0.96	.97	.110	28/	.111
Gerencia	10	" " " " 112	.113	.126	28/	.127
R1-R2	2	.128	.129	.130	30/	.131
R2-R3	2	.132	.133	.134	30/	.135

Area	IP	Mascara	Conversion	IP Router
26 Admin	10.24.0.2	255.255.255.192	10.24.0.1	
27 Ventas	.66	.224	.65	ip
28 Soporte	.98	.240	.97	
28 Gerencia	.114	.240	.113	
30 R1 FA0/0	.112	.192		
FA0/1	.1	.224		
Se2/0	.65	.224		
Se2/0	.130	.252		
Se2/0	.128			
30 R2 FA0/0	.97	.240		
Se2/0	.129	.252		
Se2/0	.134	.252		
30 R3 FA0/0	.113	.240		
Se2/0	.133	.252		
	.132			

IP route donde llegan Mascara Conimo.

Ejercicio 2: VLSM y enrutamiento dinámico OSPF en GNS3 (3 puntos) Utilizando la red base personalizada 192.B.A.0/24, diseñe e implemente el direccionamiento IP de la topología indicada por el docente. Posteriormente, configure enrutamiento dinámico mediante OSPF área 0 para permitir comunicación entre todas las redes.

Tabla de subneteo VLSM completa. Dirección de red, máscara, primer host, último host y broadcast.

192.B.A.0/24		interfaces	
B=24			
A=0			
Area	Host	IP dir	Mask
P1	40	192.24.0.0	24
P2	28	.64	27
P3	8	.96	27
P4	9	.128	28
P5	5	.144	29
R1-R2	2	.152	30
R1-R3	2	.154	30
R2-R5	2	.156	30
R3-R4	2	.158	30
R4-R5	2	.160	30

19.24.0. —

Area	IP	Mascara	Conexión
P1	.2	.192	.1
R1P2	.66 .88	.224	.88 65
P3	.98	.240	.97
P4	.130	.240	.129
P5	.146	.248	.145
R1	.129	.240	FA0/0
	.153	.252	SE2/0
	.158	.252	SE2/1
R2	.65	.224	FA0/0
	.152	.252	S2/0
	.162	.252	S2/1
R3	.97	.240	FA0/0
	.157	.252	SE2/0
	.166	.252	SE2/1
R4	.145	.248	FA0/0
	.165	.252	SE2/0
	.170	.252	SE2/1
R5	.1	.192	FA0/0
	.167	.252	SE2/0
	.169	.252	SE2/1